HANDS-ON 1 · DAY 1 / MORNING · GROUP'S SHEET

A first model on South German Credit

The case. A bank decides whether to grant consumer credits. It holds 1,000 past credits whose outcome is known, and asks whether a model could help it decide on the next applications. This sheet comes before any slide on evaluation: explore, train, look, and write down what you see. The debrief starts from your notes.

DATA CARD · SOUTH_GERMAN_CREDIT.CSV

What a row is. One consumer credit granted between 1973 and 1975 by a regional bank in southern Germany. 1,000 rows, 21 columns, no missing values.

The target. `credit_risk`: `good` if the credit contract was complied with (700 rows), `bad` if not (300 rows).

How the rows were chosen. A stratified sample: bad credits were deliberately over-represented. In the bank's portfolio about 5% of credits went bad; here 30% did. Every debtor had already passed the bank's checks: refused applicants are not in the data.

Its history. Published in 1994 as "Statlog German Credit" with a code table that mixed up the categories of eight columns (for example, it made 963 of the 1,000 debtors foreign workers; 37 were). Grömping corrected it in 2019. This file is the corrected version, with English labels.

Source and licence. U. Grömping (2019), South German Credit Data: Correcting a Widely Used Data Set, Report 4/2019, Beuth Hochschule für Technik Berlin. UCI Machine Learning Repository, dataset 573, doi:10.24432/C5QG88, CC BY 4.0.

column	what it describes	column	what it describes
`status`	the debtor's checking account with the bank	`present_residence`	years at the present address (bands)
`duration`	credit duration, months	`property`	the debtor's most valuable property
`credit_history`	how earlier credits were paid back	`age`	age, years
`purpose`	what the credit is for	`other_installment_plans`	instalment plans with other providers
`amount`	credit amount in DM, rescaled (true values unknown)	`housing`	rent, own, or for free
`savings`	the debtor's savings (bands)	`number_credits`	credits at this bank, this one included
`employment_duration`	time with the current employer (bands)	`job`	kind of job, from unskilled to manager
`installment_rate`	instalments as a share of disposable income	`people_liable`	persons who depend on the debtor
`personal_status_sex`	sex and marital status combined; sex cannot be recovered from it	`telephone`	a landline in the debtor's name
`other_debtors`	a co-applicant or a guarantor, if any	`foreign_worker`	whether the debtor is a foreign worker

THREE ROLES, ONE SCREEN

Driver. Works Dataiku, and only Dataiku: says aloud what each screen shows before clicking on.

Mapper. Keeps this sheet: writes the answers and the three notes, and places each note on a phase of the map (BU business understanding, DU data understanding, DP data preparation, M modeling, E evaluation, D deployment).

Referee. Checks every step against this sheet and challenges every conclusion: is it true of the data, or of the tool? Speaks for the group in the debrief.

STEP 1 · EXPLORE THE DATA BEFORE MODELLING IT

1 ☐ **Load the data**
In your project, on the Flow: **+ Dataset** → **Upload your files**, choose `south_german_credit.csv`, then **Create**.

2 ☐ **Count the bad credits**
Open the dataset: **Explore**, click the header of `credit_risk`, then **Analyze**.

Share of `bad`: _____ The bank's own rate was about 5%. Why would a sample hold six times more?

3 ☐ **Set one column against the target**
Charts: chart type **Stacked bars 100%**; drag `status` to the X axis, **Count of records** to the Y axis, and `credit_risk` to the colour.

Which value of `status` carries the largest share of `bad`? _____

STEP 2 · TRAIN A FIRST MODEL ON ALL 20 COLUMNS

1 ☐ **Create the analysis**
Select the dataset, then **Lab** → **AutoML Prediction**. Target `credit_risk`. Keep **Quick prototypes**, then **Create**.

2 ☐ **Bring back five columns**
Design → **Features handling**: five rows read **Rejected, text feature** (`status`, `credit_history`, `savings`, `personal_status_sex`, `property`). Dataiku took their long labels for free text. Tick the five, set their type to **Categorical**, and switch them on.

3 ☐ **Train**
Change nothing else in **Design**. Click **Train** and wait for every model to finish.

How many models did Dataiku train? _____ By which metric does **Results** rank them? _____ Best score: _____

STEP 3 · LOOK AT THE RESULT, THEN WRITE THREE NOTES

1 ☐ **Open the best model**
Results, click the best model, then **Performance** → **Confusion matrix**. Copy the table as Dataiku labels it:

	predicted _____	**predicted** _____
actually _____		
actually _____		

Which value of `credit_risk` does Dataiku treat as **positive**? _____ Drag the **threshold** slider above the table: what changes? _____

2 ☐ **Ask which columns weigh most**
Interpretation → **Variable importance**. The first three: _____

Does your chart of step 1 agree?

3 ☐ **Write three notes**
Three things you noticed, or cannot explain yet, on any screen of this sheet. The Mapper circles the phase each one belongs to; the Referee picks the one to say first.

_____ phase: BU DU DP M E D

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☐ **Reached this line? Stop here: the Referee reports in the debrief.**

Panels 4 to 7 are for block 2, each after its slides: 4 to 6 in **Evaluating a classifier**, 7 in **Model selection and cross-validation**.

4 · MOVE THE THRESHOLD

TP: bad, refused. FP: good, refused. FN: bad, granted. TN: good, granted.

1 ☐ **Reopen the model of panel 3**
Results → **model** → **Performance** → **Confusion matrix**. Set the cut-off with the slider above the table to each value below, and copy the counts.

cut-off	TP	FP	FN	TN	precision	recall	false-positive rate
0.6							
0.4							
0.2							

Each row is a point of the ROC curve: recall up, false-positive rate across (slide 12 of **Evaluating a classifier**).

Which cut-off would you put in production, and why? _____

5 · ENTER WHAT EACH ERROR COSTS

The committee fixes the costs: granting a credit that goes **bad** costs 5, refusing a **good** one costs 1, a right decision costs 0.

1 ☐ **Set the cost matrix**
Design → **Metrics**: under threshold optimization, optimize for **Cost matrix gain**. Enter the costs as negative gains: bad predicted good → −5, good predicted bad → −1, both right decisions → 0. Train again.

Threshold Dataiku now selects (**Performance** → **Confusion matrix**): _____ The rule from the slides gives 1 / (1 + 5) ≈ 0.17. Why might the two differ?

_____

6 · READ THE GAINS CHART

On the best model: **Performance** → **Lift charts**. If analysts can only review the 20% of applicants the model finds riskiest, what share of the bad credits will they see? _____

7 · REPLACE ONE SPLIT BY FIVE FOLDS

1 ☐ **Switch to cross-validation**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, stratified. Keep two algorithms: Logistic Regression and Random Forest. Train.

algorithm	ROC AUC	±
Logistic Regression		
Random Forest		

Does the ± let you say one algorithm is better? What would you need to see?

_____